

Lecture 9: More about LTI Systems.

- Casual & Stable LTI Systems

(Based on $h[n]$ & $H(z)$)

- Output of LTI system having the Difference Equation

Reminder:

Casual System: A system that its output depends on current and past inputs not future inputs $\xrightarrow{\text{for Real-Time Signal Proc.}}$

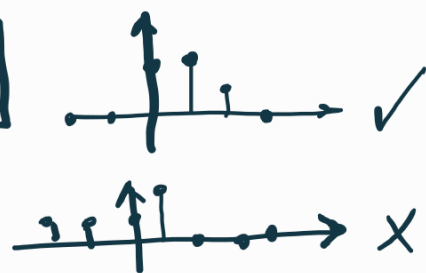
BIBO stable: A system that has bounded outputs for bounded inputs ($|x[n]| \leq m_x \rightarrow |y[n]| \leq m_y$) $\rightarrow$ to avoid damages

Now, for LTI systems having $h[n]$?

$$\begin{aligned} x[n] \rightarrow \boxed{h[n]} \rightarrow y[n] \quad y[n] &= x[n] * h[n] = h[n] * x[n] \\ &= \sum_{k=-\infty}^{+\infty} h[k] x[n-k] \quad (1) \end{aligned}$$

For causality $\rightarrow$ we don't want to have $x[n-k]$ for $k < 0$

in (1) $\rightarrow$ Hence, $\boxed{h[k] = 0 \text{ for } k < 0}$



Stability For LTI Systems:

We need to have $|y[n]| < \infty$ for $|x[n]| < \infty$:

$$\text{LTI} \rightarrow |y[n]| = \left| \sum_{k=-\infty}^{+\infty} x[k] h[n-k] \right| \leq \sum_{k=-\infty}^{+\infty} |x[k]| |h[n-k]|$$

triangle inequality

$$\stackrel{(*)}{\rightarrow} \leq M_x \sum_{k=-\infty}^{+\infty} \underbrace{|h[n-k]|}_m = M_x \sum_{m=-\infty}^{+\infty} |h[m]| < \infty$$

(**)

$$\Rightarrow \boxed{\sum_{m=-\infty}^{+\infty} |h[m]| < \infty} \rightarrow \begin{array}{l} \text{If satisfied,} \\ \text{System is BIBO} \\ \text{stable} \end{array}$$

Discussion Time 1: Which system is both casual &

BIBO stable?

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & \text{else} \end{cases}$$

Unit Step Function

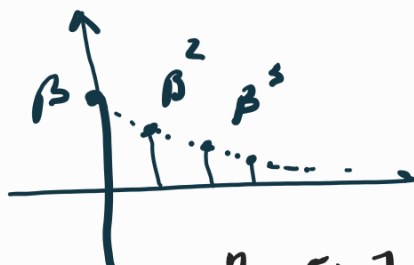
hint: if $|\beta| < 1$

$$\sum_{n=0}^{+\infty} \beta^n = \frac{1}{1-\beta}$$

(a) $h[n] = \delta[n] + \delta[n+1]$

(b) $h[n] = (1/2)^n u[-n-1]$

(c) $h[n] = (-1)^n u[n]$



(d) $(1/2)^n u[n] \checkmark \rightarrow \beta^n u[n] \quad |\beta| < 1$

Causality & stability Based on System Function $H(z)$:

↓
LTI → Rational System Function

$$H(z) = \frac{\sum_{k=0}^n b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} = \frac{B(z)}{A(z)}$$

$$= G \frac{(z - z_1) \dots (z - z_n)}{(z - p_1) \dots (z - p_N)}$$

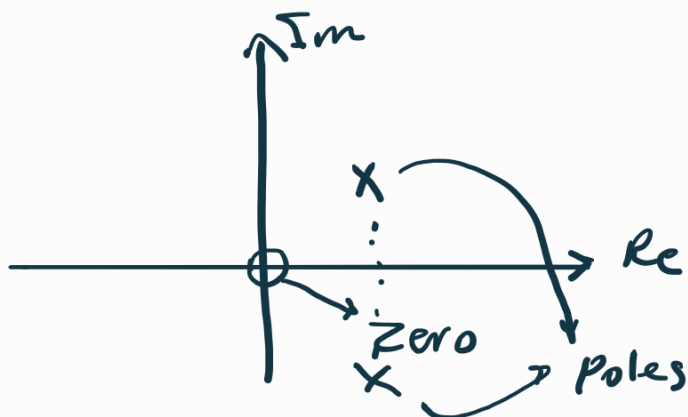
$$n' = \max\{n, N\}$$

$\left\{ \begin{array}{l} B(z_k) = 0 \rightarrow H(z_k) = 0 \rightarrow \text{It is called a Zero} \end{array} \right.$

$\left\{ \begin{array}{l} A(p_k) = 0 \rightarrow H(p_k) = \infty \rightarrow \text{" " " " Pole} \\ \downarrow \\ \text{zeros of the denominator} \end{array} \right.$

ROC cannot contain any poles as $|H(p_k)| = \infty$

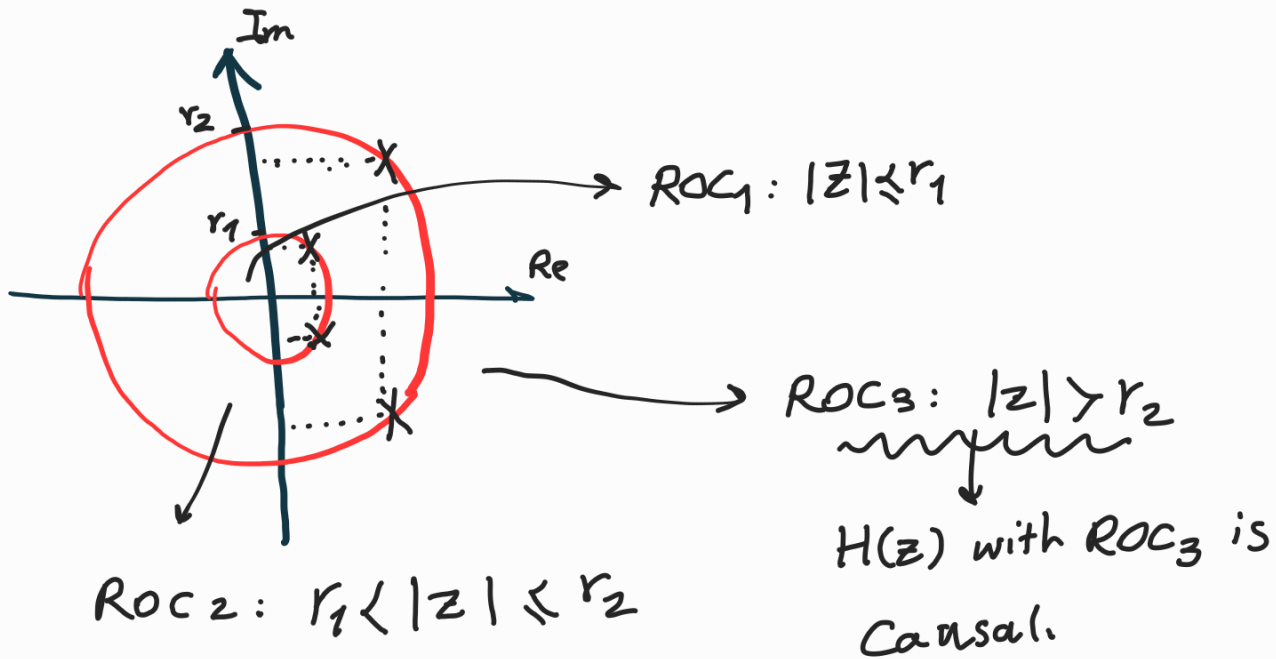
Illustration of a pole-zero plot:



Point 1: If the polynomials in the denominator and numerator have real-valued coefficients (a_k, b_k) , poles and zeros come in conjugate pairs.

Point 2: It can be shown to have a causal system, ROC should be outside some circle with radius r .
(as we need to $h[n] = 0$ for $n < 0$)

ROC: $|z| > r \rightarrow$ causal



Point 3: It can be shown that to have a BIBO stable system, unit circle ($|z| = 1$) must be included in ROC $\rightarrow$ i.e.

$$|H(z)| < \infty \text{ for } |z| = 1$$

Point 4: Causal systems are stable if all poles are inside the unit circle ($|z| = 1$).

Example 1:

$$h[n] = a^n u[n] = \begin{cases} a^n & n \geq 0 \\ 0 & \text{else} \end{cases}$$

$$H(z) = \sum h[n] z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n}$$

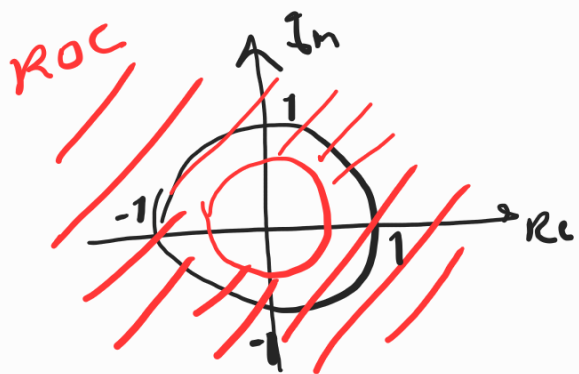
$$= \sum_{n=0}^{\infty} (a z^{-1})^n = \frac{1}{1 - a z^{-1}} \quad \text{ROC:}$$

for $|a z^{-1}| < 1 \Rightarrow |a| < |z|$

otherwise $\rightarrow \infty$

$$\sum_{n=0}^{\infty} \beta^n = \frac{1}{1 - \beta} \quad \text{for } |\beta| < 1$$

So if $|a| < 1$ (for example $1/2$)



causal & BIBO stable

just by looking at the pole

$\rightarrow \text{Pole} = a \rightarrow$ for $|a| < 1$ inside the unit circle
✓

DT2.

DT3:

$$\text{FIR} \rightarrow H(z) = G \frac{(z - z_1) \dots (z - z_m)}{z^N}$$

poles are always inside the unit circle

→ Both stable & causal

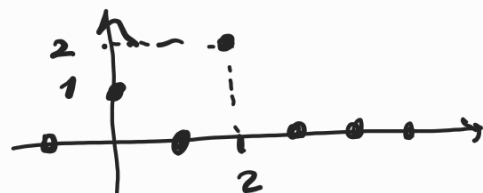
- Output of the LTI System:

$$\text{FIR Filters} \rightarrow y[n] = \sum_{k=0}^N a_k x[n-k]$$

$$\text{Example 1: } y[n] = x[n] + 2x[n-1]$$

(a) What is the impulse response of the filter?

$$x[n] = \delta[n] \rightarrow h[n] = y[n] = \delta[n] + 2\delta[n-1]$$



(b) What is the output for $x[n] = [1 \ 2 \ 3]$?
 $y[n] = 0$ for $n < 0$

$\downarrow \quad \downarrow \quad \downarrow$
 $x[0] \quad x[1] \quad x[2]$

$$y[0] = x[0] + 2x[\cancel{2}] = 1$$

$$y[1] = x[1] + 2\overset{0}{x}[\cancel{-1}] = 2$$

$$y[2] = \underset{3}{x}[\cancel{2}] + \underset{1}{2x}[\cancel{0}] = 5$$

$$y[3] = \underset{0}{x}[\cancel{3}] + 2x[1] = 4$$

$$y[4] = \underset{0}{x}[\cancel{4}] + 2x[2] = 6$$

for $n > 4 \rightarrow 0$

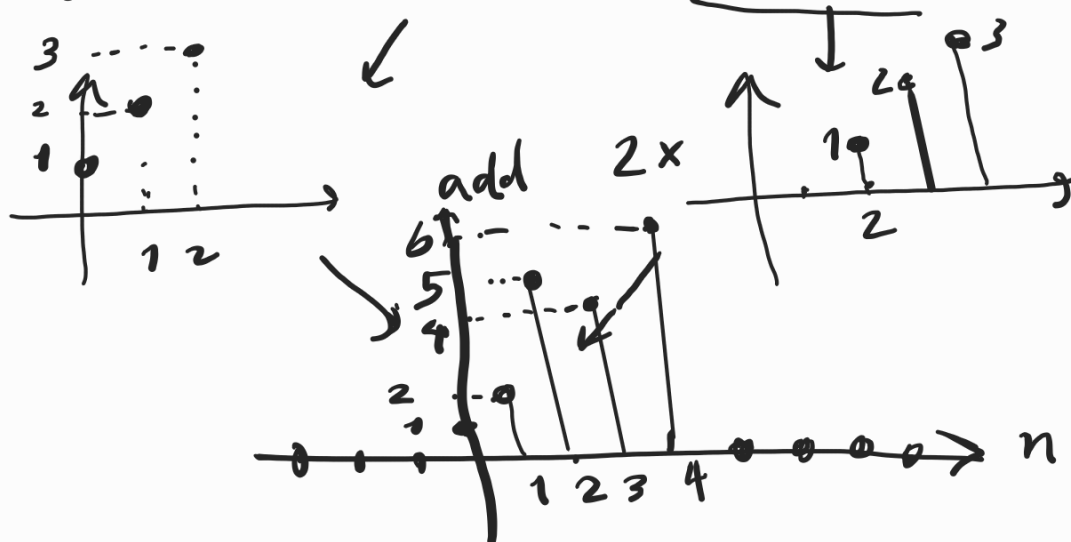
$$y[n] = [1 \ 2 \ 5 \ 4 \ 6]$$

or

$$y[n] = x[n] * h[n] = [1 \ 2 \ 3] * [1 \ 0 \ 2]$$

or

$$y[n] = x[n] + 2x[n-2]$$



IIR: $\rightarrow$ DT4. $y[n] = ?$

$$y[n] = y[n-1] + x[n] + 2x[n-2]$$

if $y[-1] = 0$ and $x[n] = [1 \ 2 \ 3]$

$$y[0] = \underset{0}{\cancel{y[-1]}} + x[0] + 2\underset{0}{\cancel{x[-2]}} = 1$$

$$y[1] = \underset{0}{y[0]} + x[1] + 2\underset{0}{\cancel{x[-1]}} = 3$$

$$y[2] = \underset{3}{\cancel{y[1]}} + \underset{3}{x[2]} + 2\underset{1}{\cancel{x[0]}} = 8$$

$$y[3] = y[2] + \underset{0}{\cancel{x[3]}} + 2x[1] = 12$$

$$y[4] = y[3] + \underset{0}{\cancel{x[4]}} + 2x[2] = 18$$

$$y[5] = y[4] + 0 = 18$$

So for $n \geq 4 \rightarrow y[n] = 18$

For more complicated systems or long inputs
 $\rightarrow$ Z-transform can be used

Example: We have the following LTI system

$$y[n] = \frac{1}{4} y[n-2] + x[n] - x[n-1]$$

What is the output for the input of $x[n]$?

$$(\text{Hint: } a^n u[n] \leftrightarrow \frac{1}{1 - a z^{-1}})$$

For LTI systems:

$$y[n] = x[n] * h[n] \rightarrow Y(z) = H(z) \cdot X(z) \quad (3)$$

$$y[n] = z^{-1} \{ Y(z) \}$$

$$x[n] = u[n] \leftrightarrow X(z) = \frac{1}{1 - z^{-1}} \quad (1)$$

$$\text{From DE} \rightarrow Y(z) = \frac{1}{4} z^{-2} Y(z) + X(z) - z^{-1} X(z)$$

z -trans.
linearity & shift
properties

$$\rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{1 - z^{-1}}{1 - \frac{1}{4} z^{-2}} \quad (2)$$

$$\textcircled{3} \xrightarrow{\textcircled{1}, \textcircled{2}} Y(z) = \frac{1}{1 - \frac{1}{4} z^{-2}} = \frac{1}{(1 - \frac{1}{2} z^{-1})(1 + \frac{1}{2} z^{-1})}$$

$$\text{Partial Fraction Decomposition} = \frac{A}{1 - \frac{1}{2} z^{-1}} + \frac{B}{1 + \frac{1}{2} z^{-1}}$$

$$A + \frac{1}{2}Az^{-1} + B - \frac{B}{2}z^{-1} = 1$$

$$\begin{cases} A+B=1 \\ A=B \end{cases} \rightarrow A=B=\frac{1}{2}$$

$$y[n] = z^{-1} \{ Y(z) \} = z^{-1} \left\{ \frac{\frac{1}{2}}{1 - \frac{1}{2}z^{-1}} + \frac{\frac{1}{2}}{1 + \frac{1}{2}z^{-1}} \right\}$$

$$\text{We know } \rightarrow a^n u[n] \leftrightarrow \frac{1}{1 - az^{-1}}$$

$$\text{Hence } \rightarrow \begin{cases} \left(\frac{1}{2}\right)^n u[n] \leftrightarrow \frac{1}{1 - \frac{1}{2}z^{-1}} \\ \left(-\frac{1}{2}\right)^n u[n] \leftrightarrow \frac{1}{1 + \frac{1}{2}z^{-1}} \end{cases}$$

$$\Rightarrow y[n] = \frac{1}{2} \left[\left(\frac{1}{2}\right)^n + \left(-\frac{1}{2}\right)^n \right] u[n]$$